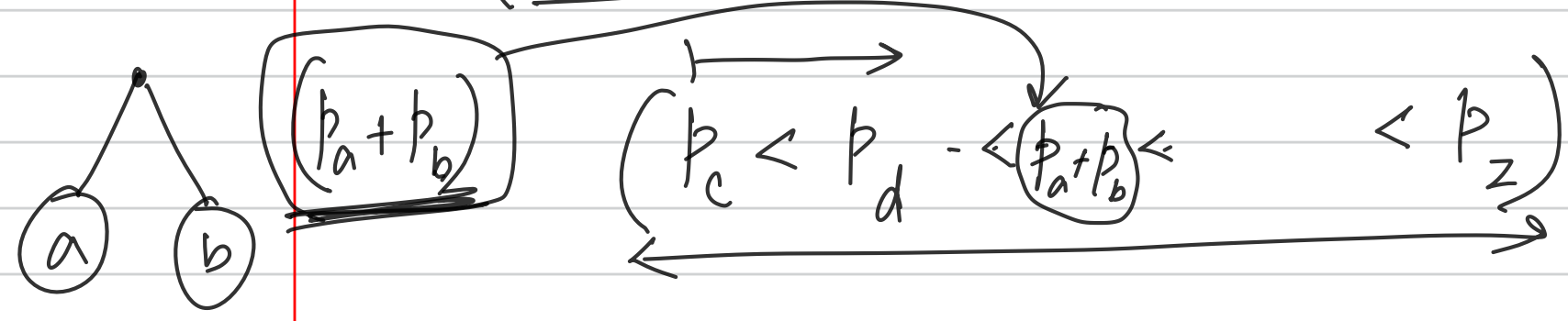
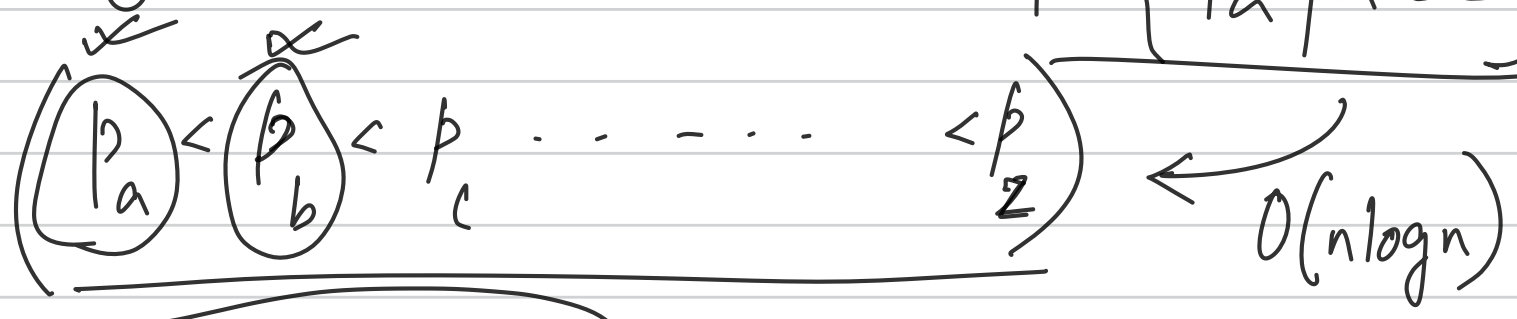
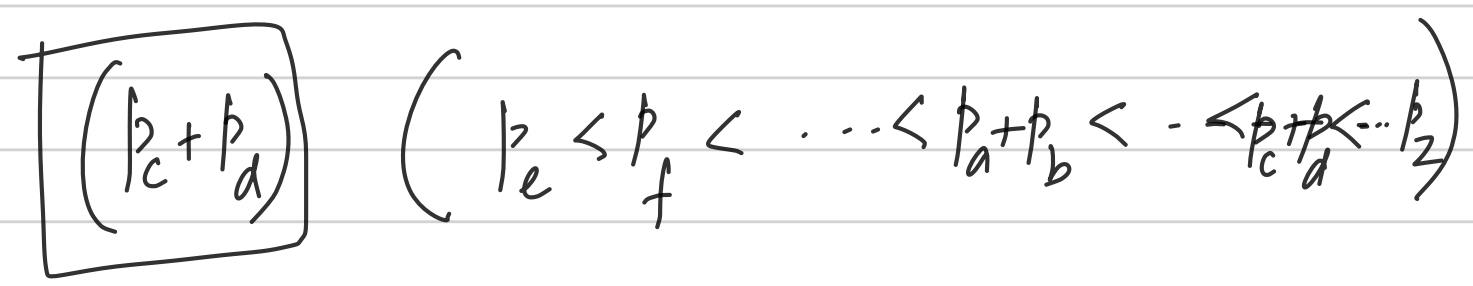


Running time: $O(n \log n)$

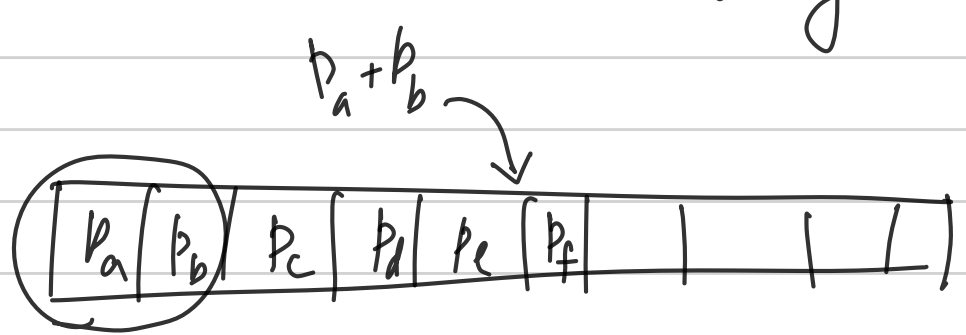
$$P = \{p_a \mid a \in \Sigma\}$$



linked list

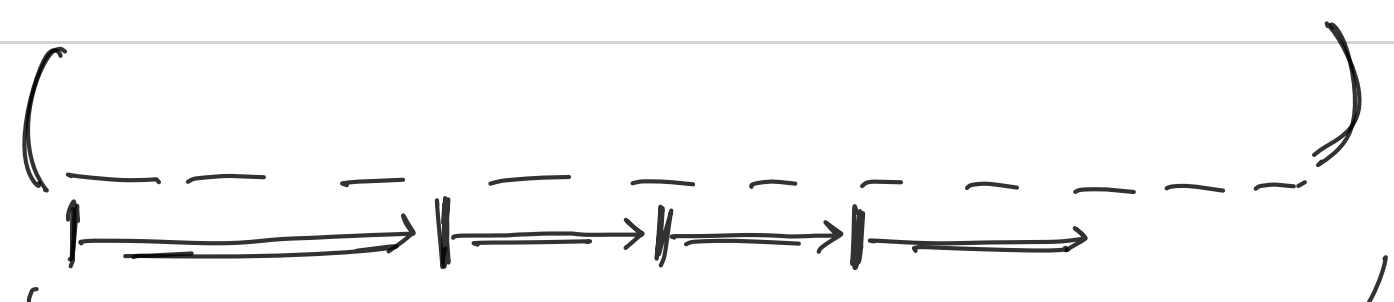


Maintaining the sorted order takes $O(n)$ time.
overall.



- ① Sorting $\underline{O(n \log n)}$
 - ② while {
- $\updownarrow$ $\underline{(n-1) * O(n)}$
- } Running time = $O(n \log n)$

(*) In some cases, if sorting can be done in $O(n)$ time then the whole algo can be implemented in $O(n)$ time.



$$O(n)$$

Th. For every alphabet Σ and non-ve symbol frequencies $\{p_a\}_{a \in \Sigma}$, the Huffman algo outputs a Σ -tree with min avg leaf depth.
 $\uparrow$
 over all possible Σ -trees.

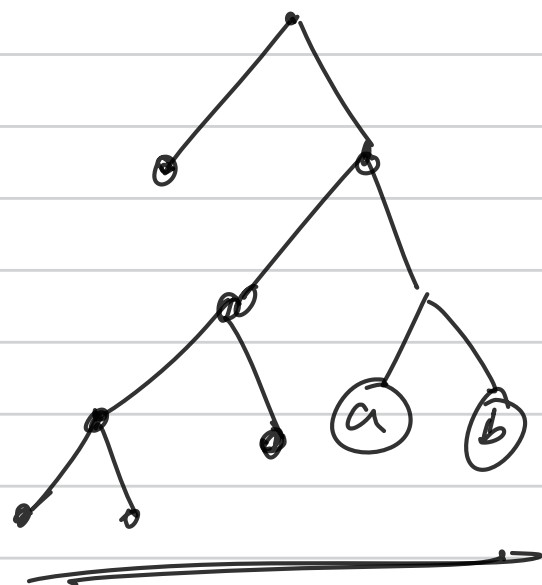
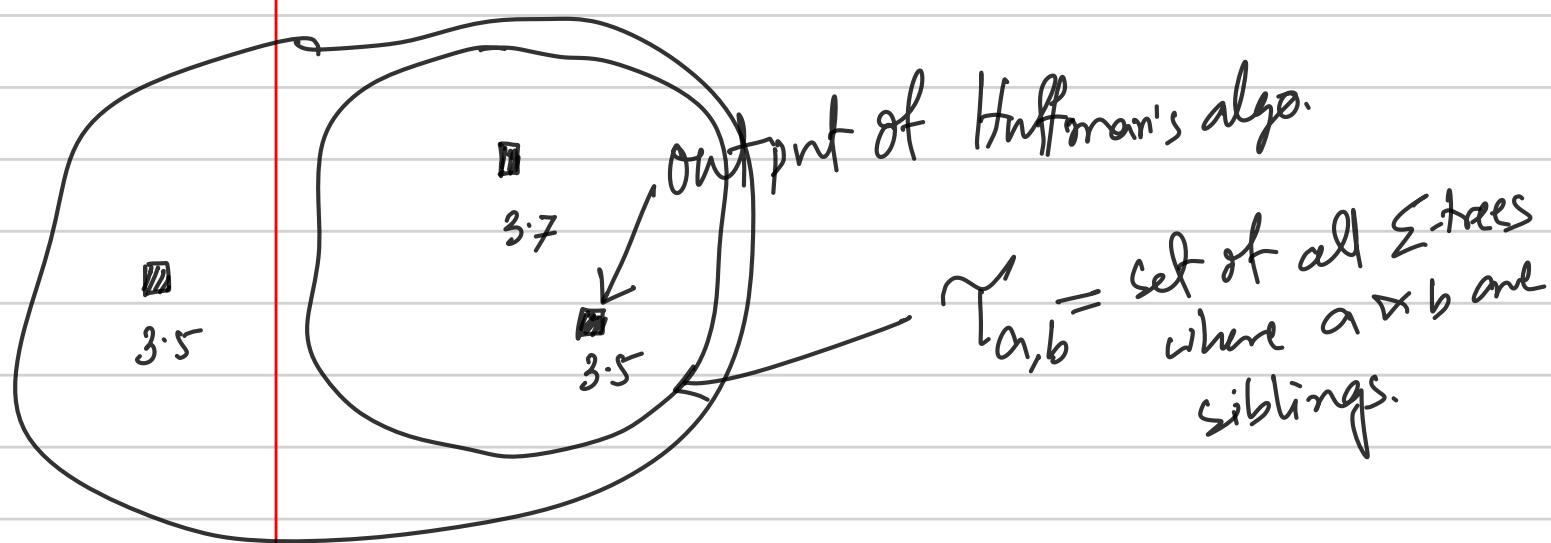
Base Case $n=2$

Ind hyp True $(n-1)$

Ind. step $|\Sigma|=n$ Claim 1 $\square$

By induction on size of alphabet set. (Σ)

Claim 1 Let $a, b \in \Sigma$ be two symbols that has min & 2nd min freq. Huffman's algo outputs a Σ -tree with min avg leaf depth over all possible Σ -trees in which a & b are siblings.



$T \Rightarrow$ set of all Σ -trees.

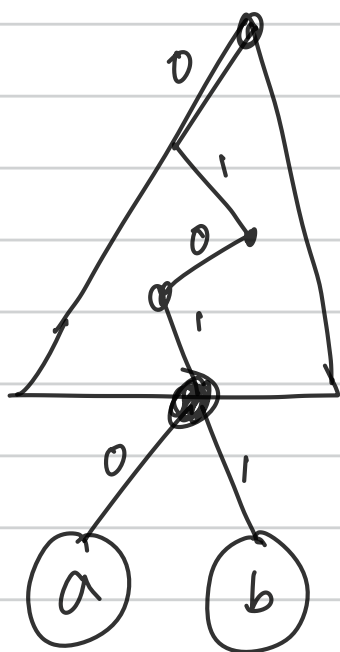
Claim 2 There is an optimal Σ -tree in $T_{a,b}$.

$$L(T, p) = \sum_{a \in \Sigma} p_a \times (\text{leaf dept of } a \text{ in } T)$$

Proof of Claim 1

$$\begin{array}{c|c} \Sigma = \{a, b, c, d, \dots\}^{(n)} & \Sigma' = \{ab, c, d, \dots\}^{(n-1)} \\ p = \{p_a, p_b, p_c, p_d, \dots\} & p' = \{p_a + p_b, p_c, p_d, \dots\} \end{array}$$

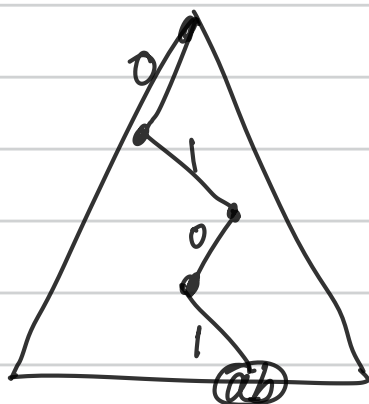
T



$$\underline{\Sigma} \times p_a + \underline{\Sigma} \times p_b$$

$$L(T, p) = L(T', p') + (p_a + p_b)$$

T'



$$4 \times p_{ab} = 4p_b + 4p_a$$

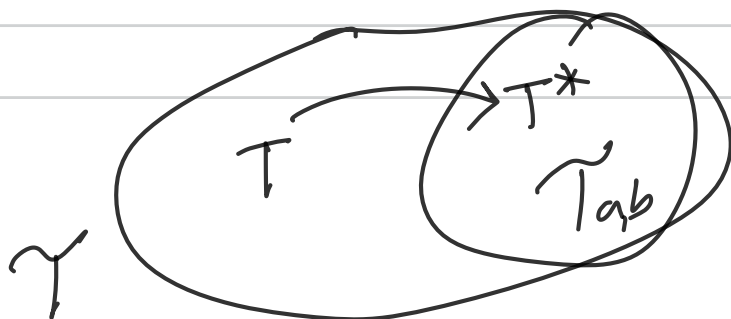
from Huffman algo. T' is the optimal Σ' -tree (By ind hyp)



T (Note that in T a & b are siblings)

$\Rightarrow$ T is the optimal

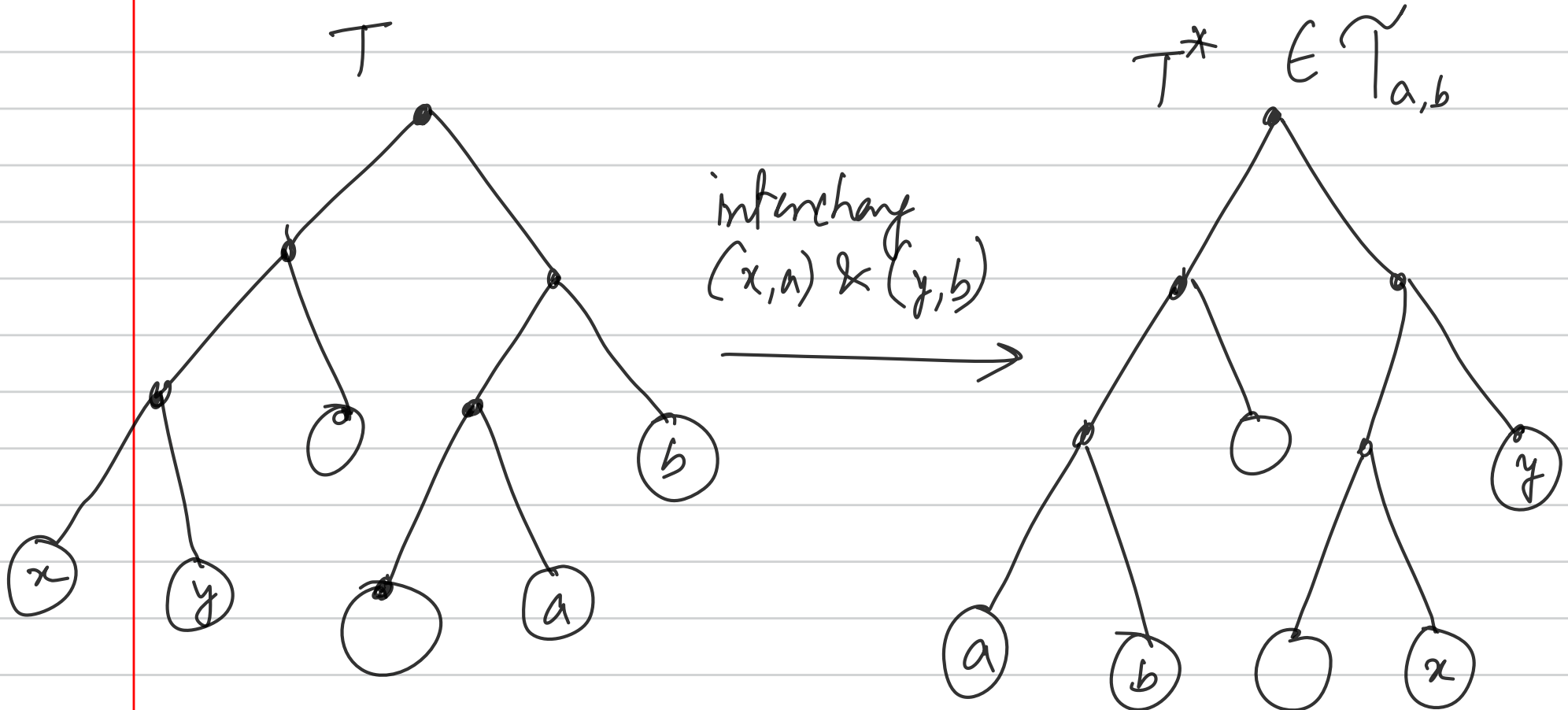
$\Rightarrow$ Huffman's algo gives optimal Σ -tree where a & b are siblings.



Proof of Claim ②

Let T be any Σ -tree. We construct $T^* \in \mathcal{T}_{a,b}$ ^{outside $\mathcal{T}_{a,b}$}
 s.t. $L(T^*, p) \leq L(T, p)$

Let (x, y) be two siblings at max depth in T .



$$\begin{aligned}
 & L(T, p) - L(T^*, p) \\
 &= \sum_{t \in \{a, b, x, y\}} p_t \times (\text{depth of } t \text{ in } T - \text{depth of } t \text{ in } T^*) \\
 &= p_a (d_T(a) - d_T(x)) + p_b (d_T(b) - d_T(y)) + p_x (d_T(x) - d_T(a)) \\
 &\quad + p_y (d_T(y) - d_T(b)) \\
 &= \underbrace{(d_T(x) - d_T(a))}_{\geq 0} \underbrace{(p_x - p_a)}_{\geq 0} + \underbrace{(d_T(y) - d_T(b))}_{\geq 0} \underbrace{(p_y - p_b)}_{\geq 0} \\
 &\geq 0
 \end{aligned}$$

$$\Rightarrow L(T, p) \geq L(T^*, p)$$

That completes the proof of Claim ②.